THE LECA RESURRECTION PROTOCOL

Recovery of the Last Eukaryotic Common Ancestor from Living Developmental Programs — One-Page Summary

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Every eukaryotic organism carries the Last Eukaryotic Common Ancestor (~1.5 Ga) as the base state of its developmental program. Arresting that program before the first multicellular step produces a LECA-grade organism. The same result from an animal, a plant, and a fungus is morphologically identical. Pre-registered 2026-03-12 with zero biological contradictions. The experiment costs $200–400 and takes days.

The Theorem

The developmental trajectory from zygote to adult IS the evolutionary trajectory of the lineage, compressed into developmental time (the Reproduction Theorem). Therefore:

E **Evolutionary grade** — the attractor the lineage occupied at that point in its history.

D **Developmental position** — the equivalent stage in the developmental program today.

$E \equiv D$ (compressed). Arresting at position D stabilises grade E. The LECA sits at $D = 0$: before the first multicellular step, in every eukaryote alive. **Arrest gives access. The information was never lost.**

Pre-registration DOI: 10.5281/zenodo.18986790

The Three Pre-Registered DOIs

Document	DOI
Pre-Registration v1.0	...18986790
Registered Amendment A1	...18989573
The Ribosome Is the LUCA	...18988844

Repository: github.com/Eric-Robert-Lawson/ attractor-oncology

The Five Pre-Registered Predictions

P1 Developmental arrest of any eukaryote before multicellular organisation produces a single-cell, nucleated, motile-capable organism.

P2 Arrested organisms from animal, plant, and fungal starting material are *morphologically indistinguishable* under light microscopy.

P3 They are *genomically distinct* by sequencing — the same attractor, different substrates.

P4 Transcriptomics: LECA-grade conserved genes active; all lineage-specific post-LECA genes epigenetically silenced.

P5 The blind panel cannot identify starting kingdom from morphology alone (above 1/3 chance).

The Three Experimental Arms

Arm	Start	Cost	Timeline
A — Fungal	*S. cerevisiae* spores	$200–350	24–72 h
B — Animal	Mouse zygote	$400–600	24–72 h
C — Plant	*Arabidopsis* zygote	$200–400	48–72 h

Full cross-kingdom run (A+B+C): $800–1,350. Arm A alone confirms P1. All three confirm P2–P5. **Arm A is executable at any BSL-1 community biology laboratory today.**

Arm A — What You Actually Do

Day 0: Sporulate *S. cerevisiae* in 1% KAc medium, 25 °C, 48–72 h. Confirm asci by microscopy.

Day 3: Wash spores into arrest medium (10 mM NaCl, 5 mM KCl, 2 mM $MgCl_2$, 1 mM $CaCl_2$, 0.1 mM $FeSO_4$, 1 mM KH_2PO_4, 0.1% glucose, **no nitrogen source**, pH 6.5–7.0). Culture at 20–22 °C, <3% O_2.

Days 4–10: Observe at 24 h, 48 h, 72 h. Photograph by phase contrast (40×). DAPI stain to confirm nuclear integrity. Check: single-cell form maintained, no germ tube, viable (≥80%).

Pass criterion: ≥80% single-cell, nucleus confirmed, ≤10% germination, viable at 72 h.

Equipment: Laminar flow cabinet, incubator, compound microscope (40× phase contrast), fluorescence microscope for DAPI. All standard BSL-1.

What the Experiment Establishes

One arm confirms	Three arms confirm
Arrest accesses LECA grade	Cross-kingdom convergence
Protocol is executable	Attractor is real
P1 confirmed	P1–P5 confirmed
Pre-reg validated (P1)	Pre-reg fully validated
Publishable result	Historic result

The oldest recoverable organism is the easiest to recover: the LECA sits at developmental position zero. **No ancient DNA. No special equipment. No institutional affiliation required.** The ribosome's peptidyl transferase centre — the LUCA's continuously executing Row 3 identity (DOI ...18988844) — will be present in every organism this protocol recovers, regardless of starting kingdom.

The geometry came first. The developmental program never lost it. The experiment stops the clock and looks.